

Supplementary Material

Protocol-Defined Semantic Fusion for Heterogeneous and Incomplete Source Coordinates

1. Reproducibility Details

The code package contains a lightweight NumPy implementation. The primary entry point is:

```
python -m qprotohop
```

The runner supports synthetic datasets, local `.npz` datasets with `x/y`, `x_train/y_train/x_test/y_test`, or `xtr/ytr/xte/yte` keys, and optional torchvision exports for MNIST and Fashion-MNIST.

2. Real-Data Commands

The real-data experiments in this manuscript were generated with:

```
scripts/run_real_cifar4_final.ps1
scripts/run_real_cifar4_shot_sweep.ps1
scripts/run_real_cifar4_comm_budget.ps1
scripts/run_real_mnist_fashion_final.ps1
scripts/run_real_mnist_fashion_hop_ablation.ps1
```

Tables and figures are produced by:

```
scripts/report_real_datasets.py
scripts/report_real_cifar4.py
scripts/plot_real_figures.py
scripts/export_real_latex_tables.py
```

3. Synthetic Commands

The synthetic schema, high-order, and communication experiments are generated by:

```
scripts/run_schema_triptych.ps1
scripts/run_hop_ablation.ps1
scripts/run_iso_communication.ps1
scripts/run_comm_budget_curve.ps1
scripts/run_noise_heterogeneity_sweep.ps1
scripts/run_schema_mismatch_sweep.ps1
```

4. Quantum-Backend and Strong-Baseline Commands

The PennyLane/Qiskit, CHOP, and stronger-baseline checks are generated by:

```
scripts/run_qproto_masked_followups.ps1
scripts/run_qproto_masked_hop_followups.ps1
scripts/run_qproto_chop_followups.ps1
scripts/run_chop_k_sweeps.ps1
scripts/run_chop_key_policy_controls.ps1
scripts/run_strong_baselines.ps1
scripts/run_schema_head_baselines.ps1
scripts/run_drift_baselines.ps1
scripts/run_quantum_param_vqc_baseline.ps1
scripts/run_if_fusion_controls.ps1
scripts/run_if_scientific_fusion.ps1
scripts/run_scale_overlap_sweeps.ps1
scripts/run_hardware_ready_demo.ps1
scripts/run_qiskit_aer_noise_sweep.ps1
```

The paired significance/effect-size table is generated from the shared summary file with:

```
scripts/report_if_significance.py
```

The real non-quantum IF benchmarks are generated by:

```
scripts/download_if_real_datasets.ps1
scripts/download_mhealth_dataset.ps1
scripts/download_hydraulic_dataset.ps1
scripts/download_pamap2_dataset.ps1
scripts/run_uci_har_if_benchmark.ps1
scripts/run_mfeat_if_benchmark.ps1
scripts/run_mhealth_if_benchmark.ps1
scripts/run_wdbc_if_benchmark.ps1
scripts/run_gas_drift_if_benchmark.ps1
scripts/run_hydraulic_if_benchmark.ps1
scripts/run_pamap2_if_benchmark.ps1
scripts/screen_gas_chop_configs.py
scripts/export_adaptive_chop_ablation.py
scripts/report_adaptive_chop_gate_choices.py
```

5. Implementation Notes

The HOP branch is applied to schema-adapted readout values rather than to the value-and-mask vector. This avoids turning missingness masks into high-order class features. The low-order branch still receives both values and masks, so missing observables remain distinguishable from zero-valued expectations.

The reported communication is computed as the number of transmitted scalar statistics times four bytes. The count includes class prototypes, class counts, high-order sketches when used, and small metadata fields. Public schema definitions and anchors are treated as protocol setup objects rather than per-round payloads.

6. Additional Experiment Tables

This appendix keeps QFL-specific and diagnostic results out of the IF main narrative while preserving the evidence needed for reproducibility and specialist review.

6.1. *Claim-to-Evidence Map*

6.2. *Protocol Manifest*

The seed counts in Table 2 are the counts used for the reported means and paired tests. Strict equal-byte tables below use the same seed identifiers, payload formulas, and public-anchor splits as the corresponding main-text rows. Random-key controls are regenerated from the experiment seed only and do not use validation or test statistics for key selection.

6.3. *Four-Class Image Pilots*

6.4. *FL and QFL Baselines*

These tables compare the protocol against FedAvg/FedProx/FedAdam, SCAFFOLD/FedDyn, FedProto, and trainable PennyLane VQC parameter aggregation. They are important specialist baselines, but they address QFL optimization rather than the central IF claim, so they are kept in the appendix.

Table 1: How the supplementary evidence supports the main scoped claims. The table is intended to make clear which results are headline evidence and which results are boundary checks.

Scoped claim	Primary evidence	Boundary or caveat
Semantic comparability is necessary before aggregation	No-schema, forced-canonical, wrong-schema, zero-fill, mask-only, shared-view, and schema-correct rows in the main image and IF-control tables	Shared-view can be strong when the common intersection already carries the signal; this is treated as a boundary case, not a failure of the protocol view
Compact value/count fusion is useful under incomplete but identified sources	UCI HAR, Multiple Features, MHEALTH, PAMAP2, Hydraulic Systems, and WDBC equal-byte tables; paired tests in Table 23	Full-dimensional HeMIS, MLP, imputation, and late-fusion rows are stronger boundary references when centralized feature-level training is allowed
HOP/CHOP is conditional, not universal	Covariance-sensing, MHEALTH, PennyLane high-order readout, CHOP sweeps, and key-policy diagnostics	HAR, MFeat, PAMAP2, WDBC, and Gas Drift show that low-order CProto or adaptive fallback is safer when the signal is low-order or imputation-friendly
Selected keys should be tested against random compression	Random-key CProto/HOP controls in MFeat, MHEALTH, WDBC, and CHOP key-policy tables	MFeat random-key low-order CProto is strong; that dataset supports schema-aware low-order fusion rather than selected-key superiority
Quantum-readout compatibility is engineering evidence, not hardware validation	Qiskit Aer count-derived robustness sweep, PennyLane high-order readouts, and hardware-ready path table	No paid IBM hardware data are used; hardware validation remains future work
Communication fairness is explicit	Strict equal-byte tables, payload accounting notes, and protocol manifest	Public schema definitions and anchors are treated as setup objects rather than per-round payloads

6.5. Scale, Overlap, and Backend Pilots

The Qiskit circuit/count-to-readout path builds one circuit per observable and converts Aer bit counts into the same precomputed-readout format used by QProto. The robustness sweep reports five seeds per noise setting; the paired optimizer/backend comparisons use the three matched seeds available across all compared QFL baselines. Both are engineering compatibility checks, not hardware validation results.

6.6. Additional Real Missing-View Fusion Check

The first two tables isolate equal-byte rows from the UCI HAR and UCI Multiple Features main-text tables. The MHEALTH table gives the full wearable multi-source result summarized in the main text. PAMAP2 is included

Table 2: Protocol manifest for the main experiments. Anchors are public/unlabeled and are used only for preprocessing, imputation, or key selection; test labels are never used for schema construction or key selection.

Experiment	Seeds	Clients	Source dictionary	Client schema sampling	Public anchors	Tuned hyperparameters
Full-chest images	5	20	simulated observable dictionary from image features	heterogeneous observable subsets/oxide	train-only public subset	K, h, λ fixed by preliminary grid
UCI HAR	5	20	feature-name sensor groups	1 common group + random private groups	train-only anchor features	PCA rank, K, h, λ , torch epochs fixed before reporting
UCI Multiple Features	10	20	six feature families	1 common weak view + 2 private views	train-only anchor features	source-balanced score, K, h, λ , torch epochs fixed before reporting
MHEALTH	10	20	chest/EKG/ankle/arm wearable sources	subject-held-out windows, 1 common source + private sources	train-only anchor windows	K, h, λ , window size, and torch epochs fixed before reporting
PAMAP2	5	20	heart-rate/hand/chest/ankle wearable sources	subject-held-out windows, 1 common source + private sources	train-only anchor windows	K, h, λ , adaptive margin, window size, and torch epochs fixed before reporting
Hydraulic Systems	5	20	named industrial sensor blocks	cohort-state clients with shared/private sensor groups	train-only anchor records	K, h, λ , HeMIS/MLP epochs, and equal-byte payload fixed before reporting
WDBC	10	20	ten biomedical measurement families	1 common family + random private families	train-only anchor features	same non-quantum grid as HAR/MFeat
Gas Drift boundary check	5	20	gas-sensor blocks	random and temporal source splits	train-only anchor records	cross-sensor HOP screen and low-order baselines fixed before reporting
Conformance sensing	5	20	controlled source dictionary	heterogeneous low-overlap schemes	generated public anchors	K, h, λ selected by communication budget
Qiskit/PennyLane checks	3-5	20	circuit-generated observable readouts	heterogeneous basis/subset/depth settings	circuit-generated public anchors	no test-label training, Aer/PennyLane settings fixed

as a larger wearable boundary case where adaptive CHOP falls back to low-order CProto. Hydraulic Systems adds a real industrial condition-monitoring check: compact CProto/CHOP statistics are far above shared-view compression and close to PCA/ridge/late-fusion baselines, while HeMIS and mask-aware MLP remain stronger full-dimensional trainable references. WDBC is included as a biomedical breadth check. It is nearly saturated, so it is less diagnostic than UCI HAR, Multiple Features, MHEALTH, or Hydraulic Systems; the main text therefore treats it as a boundary case rather than a headline comparison. Gas Drift is reported as a negative boundary check: random and temporal splits both show that CHOP is not naturally favored there, even after cross-sensor HOP pair selection and a small HOP-weight screen.

Stability and claim boundaries.. The seed-level tables should be read with the scoped claims in Table 1. The most stable comparisons are the schema-correct rows against no-schema, wrong-schema, shared-view, and equal-byte RFF controls. UCI HAR, MHEALTH, and Multiple Features also show stable advantages over equal-byte shared-view compression. Some comparisons are intentionally not used as headline claims: MFeat random-key CProto is strong, so that benchmark supports schema-aware low-order fusion rather than selected-key superiority; adaptive CHOP on MHEALTH has the highest mean among compact equal-byte statistics but is reported as interaction-heavy evidence rather than as a universal HOP guarantee; and Qiskit Aer rows are compatibility checks with limited matched seeds, not hardware validation.

Table 3: Real image datasets under heterogeneous quantum readout. Results are means with 95% confidence intervals over 5 seeds.

Dataset	Method	Acc. (95% CI)	Bal. acc.	Bytes
MNIST-4	QProto-HOP	0.709 ± 0.054	0.707	4628
	QProto prototype	0.709 ± 0.054	0.707	3092
	No schema	0.441 ± 0.029	0.437	3092
	Forced canonical	0.437 ± 0.039	0.434	3092
	Wrong schema	0.280 ± 0.023	0.278	4628
	FedAvg forced	0.374 ± 0.078	0.368	3088
	FedProx forced	0.374 ± 0.079	0.368	3088
	Classical kernel	0.671 ± 0.026	0.667	3092
Fashion-4	QProto-HOP	0.746 ± 0.037	0.746	4628
	QProto prototype	0.745 ± 0.038	0.745	3092
	No schema	0.431 ± 0.024	0.430	3092
	Forced canonical	0.429 ± 0.022	0.428	3092
	Wrong schema	0.299 ± 0.027	0.298	4628
	FedAvg forced	0.336 ± 0.064	0.339	3088
	FedProx forced	0.335 ± 0.064	0.338	3088
	Classical kernel	0.663 ± 0.017	0.663	3092
CIFAR-4	QProto-HOP	0.486 ± 0.052	0.485	4628
	QProto prototype	0.486 ± 0.052	0.485	3092
	No schema	0.339 ± 0.017	0.339	3092
	Forced canonical	0.339 ± 0.019	0.339	3092
	Wrong schema	0.268 ± 0.015	0.267	4628
	FedAvg forced	0.298 ± 0.046	0.299	3088
	FedProx forced	0.298 ± 0.046	0.299	3088
	Classical kernel	0.411 ± 0.010	0.411	3092

6.7. High-Order and Key-Policy Diagnostics

6.8. Schema, Shot, and Communication Diagnostics

7. Additional Theory Details

7.1. Plug-In Classification Consistency

Let $G(z) = g(A_S(r(x)))$ denote the protocol feature used by the classifier, where g may contain low-order RFF features, coverage-aware values, and HOP features. Define the population class statistic

$$\theta_c^* = \mathbb{E}[G(z) \mid y = c] \quad (1)$$

and the population score

$$s_c^*(z) = \|G(z) - \theta_c^*\|_2^2. \quad (2)$$

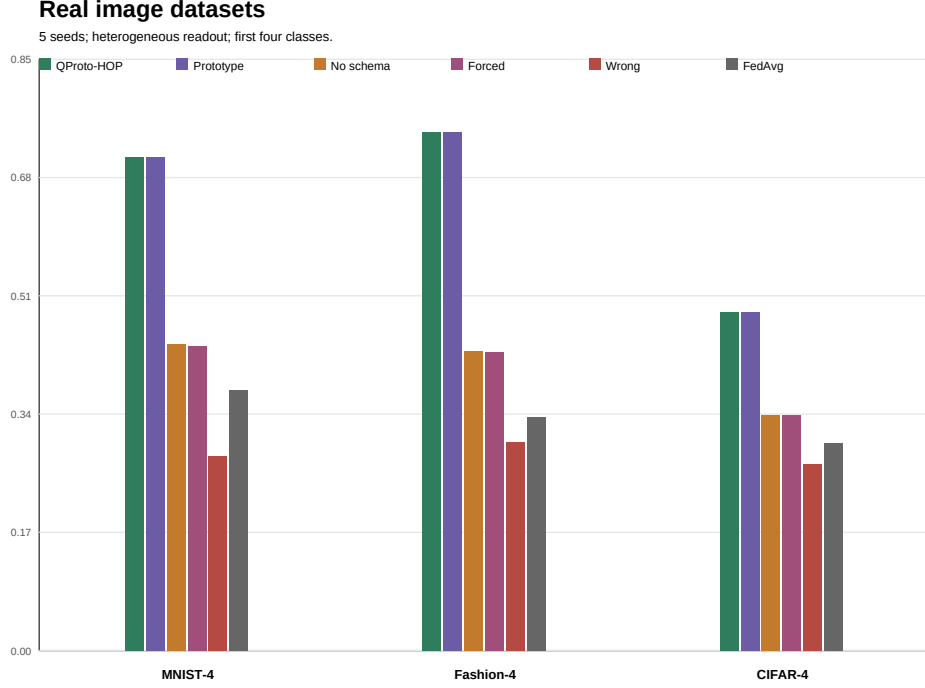


Figure 1: Four-class image diagnostics under heterogeneous quantum readout. These results motivated the full-class experiments in the main text.

The empirical server statistic $\hat{\theta}_c$ is a plug-in estimator of θ_c^* . If the margin

$$\Delta(z) = \min_{c \neq y} \{s_c^*(z) - s_y^*(z)\} \quad (3)$$

is positive and

$$\max_c |\hat{s}_c(z) - s_c^*(z)| < \Delta(z)/2, \quad (4)$$

then the empirical classifier returns the same label as the population nearest-prototype classifier. This is the margin condition used implicitly in the finite-budget theorem in the main paper. It converts estimation error in comparable readout statistics into a classification guarantee.

Table 4: Strong FL baselines under the same heterogeneous-readout protocols. Full-class rows use five seeds; Qiskit Aer and PennyLane rows use three seeds. Bytes are per client per round.

Setting	Method	Acc.	95% CI	Bytes
MNIST-10	FedAdam forced	0.229	0.011	7720
MNIST-10	FedAdam schema	0.456	0.033	7720
MNIST-10	FedProto schema	0.412	0.029	7724
MNIST-10	CProto, $K = 96$	0.738	0.030	7724
MNIST-10	CHOP, $K = 96$	0.739	0.030	11564
MNIST-10	QProto-Masked-HOP	0.744	0.027	24364
Fashion-10	FedAdam forced	0.254	0.019	7720
Fashion-10	FedAdam schema	0.544	0.033	7720
Fashion-10	FedProto schema	0.522	0.037	7724
Fashion-10	CProto, $K = 96$	0.660	0.027	7724
Fashion-10	CHOP, $K = 96$	0.660	0.027	11564
Fashion-10	QProto-Masked-HOP	0.662	0.026	24364
CIFAR-10	FedAdam forced	0.166	0.012	7720
CIFAR-10	FedAdam schema	0.281	0.017	7720
CIFAR-10	FedProto schema	0.268	0.019	7724
CIFAR-10	CProto, $K = 96$	0.450	0.005	7724
CIFAR-10	CHOP, $K = 96$	0.451	0.004	11564
CIFAR-10	QProto-Masked-HOP	0.454	0.004	24364
Qiskit Aer MNIST-4	FedAdam forced	0.379	0.034	528
Qiskit Aer MNIST-4	FedAdam schema	0.426	0.178	528
Qiskit Aer MNIST-4	FedProto schema	0.428	0.197	532
Qiskit Aer MNIST-4	CProto, $K = 8$	0.576	0.045	276
Qiskit Aer MNIST-4	CHOP, $K = 8$	0.579	0.043	532
Qiskit Aer MNIST-4	QProto-Masked-HOP	0.597	0.065	660
PennyLane high-order	FedAdam forced	0.359	0.037	1552
PennyLane high-order	FedAdam schema	0.317	0.021	1552
PennyLane high-order	FedProto schema	0.285	0.029	1556
PennyLane high-order	CProto, $K = 24$	0.403	0.061	788
PennyLane high-order	CHOP, $K = 24$	0.602	0.060	2324
PennyLane high-order	QProto-Masked-HOP	0.550	0.009	3476

7.2. Finite-Shot Constants

For a Pauli observable with outcomes in $\{-1, +1\}$, Hoeffding’s inequality gives

$$\Pr(|\hat{\mu}_{ia} - \mu_{ia}| > \epsilon) \leq 2 \exp(-S_i \epsilon^2 / 2). \quad (5)$$

After a union bound over D public observables, with probability at least $1 - \delta$,

$$\|\hat{v} - v\|_2 \leq \sqrt{\frac{2D \log(2D/\delta)}{S_{\min}}}. \quad (6)$$

Table 5: Drift-control FL baselines on quantum-backend readout tasks. Values are mean accuracy over three seeds; bytes are per client per round. SCAFFOLD transmits an additional control-variate update, so its communication is higher than ordinary head aggregation.

Setting	Method	Acc.	95% CI	Bytes
Qiskit Aer MNIST-4	FedAdam schema	0.426	0.178	528
Qiskit Aer MNIST-4	FedProto schema	0.428	0.197	532
Qiskit Aer MNIST-4	SCAFFOLD schema	0.357	0.191	1056
Qiskit Aer MNIST-4	FedDyn schema	0.418	0.188	528
Qiskit Aer MNIST-4	CProto	0.576	0.045	276
Qiskit Aer MNIST-4	CHOP	0.579	0.043	532
Qiskit Aer MNIST-4	QProto-Masked-HOP	0.597	0.065	660
PennyLane high-order	FedAdam schema	0.316	0.021	1552
PennyLane high-order	FedProto schema	0.285	0.029	1556
PennyLane high-order	SCAFFOLD schema	0.283	0.009	3104
PennyLane high-order	FedDyn schema	0.299	0.011	1552
PennyLane high-order	CProto	0.403	0.061	788
PennyLane high-order	CHOP	0.602	0.061	2324
PennyLane high-order	QProto-Masked-HOP	0.550	0.009	3476

Table 6: Trainable PennyLane VQC parameter-aggregation baseline. This baseline uses a shared ansatz and canonical readout, so it tests standard quantum-parameter FL rather than heterogeneous-readout prototype aggregation.

Method	Acc.	95% CI	Bal. acc.	Bytes
VQC init	0.243	0.085	0.235	128
VQC FedAvg	0.535	0.084	0.514	128
VQC FedProx	0.534	0.082	0.513	128
VQC centralized	0.520	0.063	0.500	128

If the feature map is L_g -Lipschitz and bounded by R_g , the induced score perturbation is bounded by

$$|\|\hat{G} - \theta_c\|_2^2 - \|G - \theta_c\|_2^2| \leq 2R_g L_g \|\hat{v} - v\|_2 + L_g^2 \|\hat{v} - v\|_2^2. \quad (7)$$

This explains why shot noise enters the finite-budget theorem through a $D^{1/2} S_{\min}^{-1/2}$ term, while schema mismatch does not vanish with shots.

7.3. Coverage-Aware Estimator Details

The coverage-aware estimator stores two sufficient statistics per class and source coordinate: a value sum and an observation count. This is equivalent to estimating

$$\mathbb{E}[V_j \mid y = c, M_j = 1] \quad (8)$$

Table 7: Scale and observable-overlap stress tests. Values are mean accuracy over three seeds. Detailed confidence intervals are reported in the released experiment logs.

Setting	Stress	FedAdam	FedProto	Shared	CProto	CHOP	Masked-HOP
MNIST-10	50 clients	0.518	0.532	0.702	0.745	0.747	0.752
MNIST-10	100 clients	0.553	0.584	0.701	0.746	0.748	0.753
Fashion-10	50 clients	0.590	0.602	0.660	0.669	0.669	0.672
Fashion-10	100 clients	0.614	0.620	0.660	0.670	0.671	0.673
CIFAR-10	50 clients	0.320	0.325	0.424	0.449	0.449	0.453
CIFAR-10	100 clients	0.342	0.348	0.424	0.450	0.451	0.455
MNIST-10	20% overlap	0.173	0.158	0.555	0.688	0.689	0.737
MNIST-10	100% overlap	0.681	0.719	0.718	0.744	0.745	0.745
Fashion-10	20% overlap	0.238	0.220	0.598	0.651	0.651	0.670
Fashion-10	100% overlap	0.673	0.668	0.667	0.670	0.672	0.671
CIFAR-10	20% overlap	0.148	0.143	0.325	0.424	0.425	0.453
CIFAR-10	100% overlap	0.420	0.438	0.438	0.459	0.460	0.460
PennyLane high-order	20% overlap	0.299	0.280	0.425	0.385	0.593	0.558
PennyLane high-order	50% overlap	0.314	0.291	0.475	0.433	0.662	0.616
PennyLane high-order	80% overlap	0.429	0.394	0.597	0.440	0.691	0.662

instead of estimating $\mathbb{E}[M_j V_j \mid y = c]$. Zero-fill prototypes estimate the latter quantity, which mixes measurement value and observation probability. This distinction is important when missingness is caused by source availability or measurement design: a missing observable should not be interpreted as a measured zero expectation. If missingness is conditionally independent of the value given the class, the coverage-aware estimator targets the full class-conditional coordinate mean. If missingness is informative, it targets the observed-view mean and the mask/coverage pattern remains available to the classifier.

7.4. Schema-Uncertainty Details

The main schema mismatch theorem treats a fully wrong permutation for clarity. A partially wrong schema produces the same kind of non-vanishing semantic bias. If a fraction ρ of coordinate mass is assigned to incorrect public keys, boundedness gives an $O(\rho)$ bias term in the affected class prototypes. This term is separate from sample, shot, and random-feature errors. It can only be reduced by improving schema metadata, using calibration anchors to detect mismatches, or modeling schema uncertainty explicitly. The current method assumes protocol-defined observable identities are available; noisy schema inference is left for future work.

Table 8: Quantum-backend compatibility checks after adding coverage-aware aggregation. PennyLane uses precomputed VQC readout features; Qiskit Aer uses noisy finite-shot simulator readouts.

Setting	Method	Acc.	95% CI	Bytes
PennyLane MNIST-4	QProto-Masked	0.733	0.063	1940
PennyLane MNIST-4	QProto-Masked-HOP	0.734	0.061	2708
PennyLane MNIST-4	low-order prototype	0.413	–	1556
PennyLane private-signal	QProto-Masked	0.748	0.032	1940
PennyLane private-signal	QProto-Masked-HOP	0.750	0.029	2708
PennyLane private-signal	no-schema	0.405	0.041	1556
PennyLane private-signal	shared-observable	0.357	0.013	1556
Qiskit Aer MNIST-4	QProto-Masked	0.593	0.060	404
Qiskit Aer MNIST-4	QProto-Masked-HOP	0.597	0.065	660
Qiskit Aer MNIST-4	CHOP, $K = 8$	0.579	0.043	532
Qiskit Aer MNIST-4	CProto, $K = 8$	0.576	0.046	276
Qiskit Aer MNIST-4	shared-observable	0.558	0.083	532
Qiskit Aer MNIST-4	low-order prototype	0.427	0.238	532
Qiskit Aer MNIST-4	no-schema	0.375	0.155	532

7.5. Random-Feature and Communication Terms

For bounded shift-invariant kernels, random Fourier features estimate kernel values with Monte Carlo error $O(d^{-1/2})$ for fixed pairs. A uniform statement over M evaluated pairs follows by a union bound:

$$\max_{\ell \leq M} |\hat{k}(z_\ell, z'_\ell) - k(z_\ell, z'_\ell)| = O\left(\sqrt{\frac{\log(M/\delta)}{d}}\right). \quad (9)$$

Because each class prototype uses d low-order coordinates and h high-order coordinates, the payload scales as $C(d+h)$. Thus, increasing communication reduces sketch error but does not reduce schema error. This separation is central to the protocol claim: no amount of RFF or HOP dimension can fix an incorrect observable map.

7.6. HOP Advantage Under Equal Means

Suppose two classes have equal adapted readout means but covariance difference $A = \Sigma_1 - \Sigma_2 \neq 0$. For an isotropic Gaussian projection p ,

$$\mathbb{E}_p[(p^\top A p)^2] = 2\|A\|_F^2 + (\text{tr } A)^2. \quad (10)$$

Table 9: Qiskit Aer robustness sweep with count-derived readout expectations. Values are mean accuracy over five seeds. The QProto runner adds no extra synthetic readout noise here; finite shots and noise are already present in the Aer-generated measurement counts.

Setting	Wrong	No schema	FedAdam	FedProto	Shared	CProto	CHOP	Masked-HOP
Clean, depth 1, 1024 shots	0.320	0.360	0.438	0.454	0.518	0.568	0.572	0.580
Low shot, 128 shots	0.309	0.340	0.398	0.410	0.466	0.546	0.550	0.547
High readout error	0.314	0.356	0.425	0.437	0.496	0.573	0.583	0.572
High depolarizing noise	0.312	0.352	0.431	0.444	0.506	0.585	0.593	0.593
Depth 2	0.323	0.370	0.462	0.482	0.538	0.584	0.589	0.612

Table 10: Hardware-ready Qiskit circuit path on a small MNIST-2 Aer fallback. This is not a hardware claim; it verifies the same circuit/count/readout pipeline used for IBM backend execution.

Method	Acc.	Bytes
QProto-Masked-HOP	0.778	332
CHOP	0.756	268
CProto	0.753	140
FedProto schema	0.650	268
FedAdam schema	0.528	264
No schema	0.583	268
Wrong schema	0.561	396

Therefore, whenever the class covariance matrices differ, projected quadratic features have nonzero expected squared gap even when low-order mean prototypes coincide. With h independent HOP projections, the probability that every projected gap is below a fixed detectable threshold decays exponentially in h under standard small-ball assumptions. This is the formal reason CHOP is strong on the covariance-sensing and PennyLane high-order tasks but only mildly helpful on low-order image tasks.

7.7. Group-Balanced Source Scoring

The group-balanced CProto rule is used when public coordinates are partitioned into semantic source groups $G_1, \dots, G_Q$, such as sensor blocks or feature families. Let

$$\ell_{cq}(x) = \frac{\sum_{j \in G_q} m_j(x) \mathbf{1}\{N_{cj} > 0\} w_{cj} (v_j(x) - \bar{v}_{cj})^2}{\sum_{j \in G_q} m_j(x) \mathbf{1}\{N_{cj} > 0\} w_{cj} + \epsilon} \quad (11)$$

be the within-group coverage-normalized class distance. Group-balanced scoring uses

$$d_c^{\text{grp}}(x) = |\mathcal{Q}_x|^{-1} \sum_{q \in \mathcal{Q}_x} \ell_{cq}(x), \quad (12)$$

Table 11: Strict equal-byte controls on the UCI HAR missing-view sensor benchmark. All rows use the same payload as CHOP.

Method	Acc.	Bytes
Shared-view RFF equal bytes	0.391	7708
Zero-fill RFF equal bytes	0.344	7708
CProto equal bytes	0.753	7708
Group-balanced CProto equal bytes	0.716	7708
Random-key CProto equal bytes	0.632	7708
Random-key HOP equal bytes	0.616	7708
CHOP equal bytes	0.632	7708
Adaptive CHOP equal bytes	0.752	7708

Table 12: Strict equal-byte controls on the UCI Multiple Features multi-view benchmark. All rows use the same payload as CHOP.

Method	Acc.	Bytes
Shared-view RFF equal bytes	0.694	16684
Zero-fill RFF equal bytes	0.766	16684
CProto equal bytes	0.868	16684
Group-balanced CProto equal bytes	0.876	16684
Random-key CProto equal bytes	0.891	16684
Random-key HOP equal bytes	0.883	16684
CHOP equal bytes	0.722	16684
Adaptive CHOP equal bytes	0.866	16684

where $\mathcal{Q}_x$ is the set of observed semantic groups.

Proposition 1 (Dimensional dominance of coordinate-wise scoring). *Assume each semantic group G_q has an expected within-group loss $L_{cq} = \mathbb{E}[\ell_{cq}(X)]$, and suppose a coordinate-wise distance averages over all observed coordinates without group normalization. If all coordinates in a group have comparable expected loss, the coordinate-wise score estimates a dimension-weighted objective proportional to*

$$\sum_q |G_q| L_{cq}, \quad (13)$$

whereas group-balanced scoring estimates a source-balanced objective proportional to

$$\sum_q L_{cq}. \quad (14)$$

Thus, when source groups have very different dimensionalities, coordinate-wise scoring can be dominated by the largest source even if the smaller sources

Table 13: Real wearable multi-source missing-view fusion benchmark on UCI MHEALTH under a subject-held-out split. Windowed readout statistics are built from chest accelerometer, chest ECG, ankle IMU, and arm IMU sources. Equal-byte rows use the same per-client per-round payload as CHOP.

Method	Acc.	95% CI	Bytes	Runtime (s)
Mask-only metadata	0.084	0.003	11092	0.580
Zero-fill prototype	0.789	0.024	11092	0.586
Mean-impute prototype	0.789	0.026	11092	0.705
Value+mask prototype	0.266	0.017	22132	1.229
Anchor PCA imputation	0.760	0.028	11092	11.689
Learned ridge imputation	0.828	0.028	11092	1.329
Anchor autoencoder imputation	0.804	0.029	11092	11.248
Mask-aware pooled MLP	0.860	0.020	22132	1.853
HeMIS modality-dropout fusion	0.889	0.025	22132	7.169
View-wise late fusion	0.814	0.026	11092	0.333
Shared-view prototype	0.719	0.044	1492	0.072
Shared-view RFF equal bytes	0.720	0.046	13876	0.864
Zero-fill RFF equal bytes	0.774	0.028	13876	1.005
Coverage prototype all	0.820	0.027	22132	1.406
Group-balanced CProto all	0.814	0.026	22132	1.644
CProto equal bytes	0.860	0.017	13876	0.917
Group-balanced CProto equal bytes	0.826	0.015	13876	1.034
Random-key CProto equal bytes	0.798	0.033	13876	0.625
Random-key HOP equal bytes	0.798	0.033	13876	1.209
CHOP equal bytes	0.865	0.041	13876	1.217
Adaptive CHOP equal bytes	0.878	0.022	13876	1.200

are semantically equally important.

Proof. Ignoring the small ϵ term and conditioning on the observed groups, coordinate-wise scoring is an average of per-coordinate squared losses over $\cup_q G_q$. If the average coordinate loss within group q is L_{cq} , then the numerator contributes approximately $|G_q|L_{cq}$. Summing over groups gives a dimension-weighted objective. Group-balanced scoring first averages within G_q and then averages over groups, so each observed semantic source contributes one group-level term. This is the intended objective in multi-source fusion settings where a source’s importance should not be determined only by how many exported coordinates it has. $\square$

7.8. Adaptive CHOP Calibration Bound

Adaptive CHOP chooses among a finite candidate set $\mathcal{A}$, containing the low-order CProto branch and a small grid of CHOP mixtures. The choice is made on a training-only calibration split and the selected candidate is then refit on the full training set. The following bound clarifies what this gate can and cannot guarantee.

Table 14: Strict equal-byte controls on the real UCI MHEALTH wearable multi-source benchmark. All rows use the same payload as CHOP.

Method	Acc.	Bytes
Shared-view RFF equal bytes	0.720	13876
Zero-fill RFF equal bytes	0.774	13876
CProto equal bytes	0.860	13876
Group-balanced CProto equal bytes	0.826	13876
Random-key CProto equal bytes	0.798	13876
Random-key HOP equal bytes	0.798	13876
CHOP equal bytes	0.865	13876
Adaptive CHOP equal bytes	0.878	13876

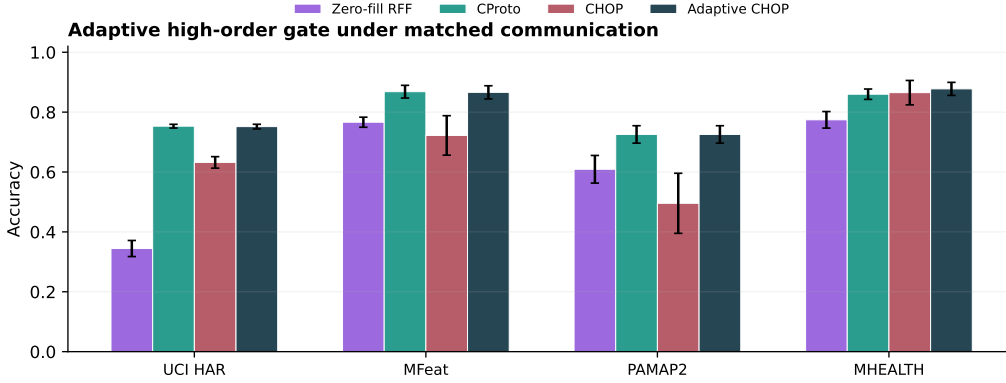


Figure 2: Adaptive CHOP summary under matched communication. The mixture/fallback gate avoids pure-CHOP degradation on low-order tasks and gives the highest mean among compact equal-byte statistics on the ten-seed MHEALTH run.

Proposition 2 (Finite calibration selection). *Let $\mathcal{A}$ contain M candidate scoring rules with true calibration accuracies $a(\alpha)$ and empirical calibration accuracies $\hat{a}(\alpha)$ computed on n_{cal} independent calibration examples. Let $\hat{\alpha} = \arg \max_{\alpha \in \mathcal{A}} \hat{a}(\alpha)$. With probability at least $1 - \delta$,*

$$a(\hat{\alpha}) \geq \max_{\alpha \in \mathcal{A}} a(\alpha) - 2\sqrt{\frac{\log(2M/\delta)}{2n_{\text{cal}}}}. \quad (15)$$

If the protocol requires a margin τ over the low-order branch before enabling CHOP, the selected high-order branch has the same guarantee relative to the best candidate in $\mathcal{A}$, while false high-order activation is controlled by the same uniform deviation term plus τ .

Table 15: Real wearable multi-source missing-view fusion benchmark on UCI PAMAP2 under a subject-held-out split. Windowed readout statistics are built from heart-rate, hand IMU, chest IMU, and ankle IMU sources. Equal-byte rows use the same per-client per-round payload as CHOP.

Method	Acc.	95% CI	Bytes	Runtime (s)
Mask-only metadata	0.081	0.007	17812	2.110
Zero-fill prototype	0.668	0.033	17812	2.175
Mean-impute prototype	0.669	0.033	17812	2.519
Value+mask prototype	0.199	0.036	35572	4.269
Anchor PCA imputation	0.693	0.049	17812	61.355
Learned ridge imputation	0.737	0.047	17812	4.293
Anchor autoencoder imputation	0.730	0.045	17812	25.229
Mask-aware pooled MLP	0.714	0.062	35572	4.836
HeMIS modality-dropout fusion	0.753	0.066	35572	15.617
View-wise late fusion	0.725	0.042	17812	1.159
Shared-view prototype	0.269	0.021	532	0.062
Shared-view RFF equal bytes	0.275	0.020	18484	2.437
Zero-fill RFF equal bytes	0.609	0.065	18484	3.158
Coverage prototype all	0.718	0.045	35572	5.158
Group-balanced CProto all	0.725	0.042	35572	6.159
CProto equal bytes	0.725	0.041	18484	2.779
Group-balanced CProto equal bytes	0.702	0.058	18484	3.171
Random-key CProto equal bytes	0.701	0.039	18484	1.794
Random-key HOP equal bytes	0.702	0.039	18484	3.425
CHOP equal bytes	0.495	0.142	18484	3.447
Adaptive CHOP equal bytes	0.725	0.041	18484	3.098

Proof. For a fixed candidate, Hoeffding’s inequality gives

$$\Pr(|\hat{a}(\alpha) - a(\alpha)| > \epsilon) \leq 2 \exp(-2n_{\text{cal}}\epsilon^2). \quad (16)$$

A union bound over M candidates gives uniform deviation at level $\epsilon = \sqrt{\log(2M/\delta)/(2n_{\text{cal}})}$. Let $\alpha^* = \arg \max_{\alpha} a(\alpha)$. Then

$$a(\hat{\alpha}) \geq \hat{a}(\hat{\alpha}) - \epsilon \geq \hat{a}(\alpha^*) - \epsilon \geq a(\alpha^*) - 2\epsilon. \quad (17)$$

The margin rule compares the best CHOP mixture to the low-order candidate using the same uniformly controlled empirical accuracies, so the probability of enabling a worse high-order branch is bounded by the same calibration uncertainty unless the true gap lies within the margin band. This is why the method treats adaptive CHOP as a train-calibrated fallback rule, not as a universal high-order guarantee. $\square$

7.9. Combined Finite-Budget Error Decomposition

The main theorems separate semantic error, sampling error, finite-shot error, sketch approximation, and high-order allocation. For a compact way

Table 16: Strict equal-byte controls on the real UCI PAMAP2 wearable multi-source benchmark. All rows use the same payload as CHOP.

Method	Acc.	Bytes
Shared-view RFF equal bytes	0.275	18484
Zero-fill RFF equal bytes	0.609	18484
CProto equal bytes	0.725	18484
Group-balanced CProto equal bytes	0.702	18484
Random-key CProto equal bytes	0.701	18484
Random-key HOP equal bytes	0.702	18484
CHOP equal bytes	0.495	18484
Adaptive CHOP equal bytes	0.725	18484

to read them together, write Π for the schema used by the server and let Π^* be the true semantic map. For a fixed class and query, the score error can be decomposed as

$$\begin{aligned}
|\hat{s}_c(z) - s_c^*(z)| \leq & \underbrace{E_{\text{schema}}(\Pi, \Pi^*)}_{\text{semantic bias}} + \underbrace{E_{\text{sample}}(n_c)}_{\text{finite examples}} + \underbrace{E_{\text{shot}}(S_{\min})}_{\text{finite readout}} \\
& + \underbrace{E_{\text{rff}}(d)}_{\text{low-order sketch}} + \underbrace{E_{\text{hop}}(h)}_{\text{high-order sketch}} + \underbrace{E_{\text{cal}}(M, n_{\text{cal}})}_{\text{adaptive gate}}.
\end{aligned} \tag{18}$$

Under the assumptions in the main text, these terms may be chosen with high probability as

$$\begin{aligned}
E_{\text{sample}} &= O(n_c^{-1/2}), \quad E_{\text{shot}} = O(D^{1/2} S_{\min}^{-1/2}), \\
E_{\text{rff}} &= O(d^{-1/2}), \quad E_{\text{hop}} = O(h^{-1/2}),
\end{aligned} \tag{19}$$

and

$$E_{\text{cal}} = O\left(\sqrt{\frac{\log M + \log(1/\delta)}{n_{\text{cal}}}}\right). \tag{20}$$

The important qualitative distinction is that E_{schema} does not vanish with more samples, shots, or sketch dimensions unless the schema itself is corrected. This is the mathematical version of the paper’s central information-fusion claim: communication and estimation can refine comparable evidence, but they cannot repair anonymous or wrongly named coordinates.

Table 17: Real industrial multi-source missing-view fusion benchmark on UCI Hydraulic Systems condition monitoring (cooler target). Each sensor time series is summarized into cycle-level readout statistics, and each client observes a heterogeneous sensor subset. Equal-byte rows use the same per-client per-round payload as CHOP.

Method	Acc.	95% CI	Bytes	Runtime (s)
Mask-only metadata	0.333	0.000	2056	0.060
Zero-fill prototype	0.940	0.030	2056	0.056
Mean-impute prototype	0.939	0.031	2056	0.072
Value+mask prototype	0.667	0.095	4096	0.129
Anchor PCA imputation	0.980	0.006	2056	2.705
Learned ridge imputation	0.978	0.007	2056	0.533
Anchor autoencoder imputation	0.980	0.007	2056	5.384
Mask-aware pooled MLP	0.993	0.009	4096	0.420
HeMIS modality-dropout fusion	0.997	0.004	4096	5.429
View-wise late fusion	0.975	0.012	2056	0.015
Shared-view prototype	0.799	0.018	256	0.004
Shared-view RFF equal bytes	0.833	0.018	3472	0.159
Zero-fill RFF equal bytes	0.968	0.014	3472	0.259
Coverage prototype all	0.972	0.018	4096	0.130
Group-balanced CProto all	0.975	0.012	4096	0.153
CProto equal bytes	0.973	0.016	3472	0.107
Group-balanced CProto equal bytes	0.973	0.012	3472	0.116
Random-key CProto equal bytes	0.969	0.024	3472	0.068
Random-key HOP equal bytes	0.967	0.025	3472	0.133
CHOP equal bytes	0.977	0.016	3472	0.134

Table 18: Strict equal-byte controls on the real UCI Hydraulic Systems benchmark. All rows use the same payload as CHOP.

Method	Acc.	Bytes
Shared-view RFF equal bytes	0.833	3472
Zero-fill RFF equal bytes	0.968	3472
CProto equal bytes	0.973	3472
Group-balanced CProto equal bytes	0.973	3472
Random-key CProto equal bytes	0.969	3472
Random-key HOP equal bytes	0.967	3472
CHOP equal bytes	0.977	3472

Table 19: Real non-quantum missing-view fusion benchmark on the WDBC biomedical diagnosis dataset. Measurement families are treated as semantic views; each client observes a heterogeneous subset. Equal-byte rows use the same per-client per-round payload as CHOP.

Method	Acc.	95% CI	Bytes	Runtime (s)
Mask-only metadata	0.509	0.015	252	0.003
Zero-fill prototype	0.921	0.006	252	0.003
Mean-impute prototype	0.922	0.006	252	0.004
Value+mask prototype	0.780	0.045	492	0.003
Anchor PCA imputation	0.882	0.014	252	0.565
Learned ridge imputation	0.924	0.008	252	0.029
Anchor autoencoder imputation	0.920	0.009	252	6.481
Mask-aware pooled MLP	0.920	0.008	492	0.706
View-wise late fusion	0.921	0.008	252	0.015
Shared-view prototype	0.896	0.009	36	0.004
Shared-view RFF equal bytes	0.896	0.006	396	0.010
Zero-fill RFF equal bytes	0.911	0.008	396	0.014
Coverage prototype all	0.922	0.006	492	0.017
Group-balanced CProto all	0.921	0.008	492	0.050
CProto equal bytes	0.922	0.006	396	0.011
Group-balanced CProto equal bytes	0.921	0.008	396	0.043
Random-key HOP equal bytes	0.905	0.011	396	0.017
CHOP equal bytes	0.923	0.006	396	0.019

Table 20: Strict equal-byte controls on the WDBC biomedical missing-view benchmark. All rows use the same payload as CHOP.

Method	Acc.	Bytes
Shared-view RFF equal bytes	0.896	396
Zero-fill RFF equal bytes	0.911	396
CProto equal bytes	0.922	396
Group-balanced CProto equal bytes	0.921	396
Random-key HOP equal bytes	0.905	396
CHOP equal bytes	0.923	396

Table 21: Strict equal-byte controls on the real Gas Sensor Array Drift benchmark. All rows use the same payload as CHOP.

Method	Acc.	Bytes
Shared-view RFF equal bytes	0.448	6940
Zero-fill RFF equal bytes	0.601	6940
CProto equal bytes	0.603	6940
Group-balanced CProto equal bytes	0.594	6940
Random-key CProto equal bytes	0.598	6940
Random-key HOP equal bytes	0.598	6940
CHOP equal bytes	0.592	6940
Adaptive CHOP equal bytes	0.604	6940

Table 22: Strict equal-byte controls on the real Gas Sensor Array Drift benchmark. All rows use the same payload as CHOP.

Method	Acc.	Bytes
Shared-view RFF equal bytes	0.247	6940
Zero-fill RFF equal bytes	0.367	6940
CProto equal bytes	0.375	6940
Group-balanced CProto equal bytes	0.398	6940
Random-key CProto equal bytes	0.376	6940
Random-key HOP equal bytes	0.377	6940
CHOP equal bytes	0.364	6940
Adaptive CHOP equal bytes	0.364	6940

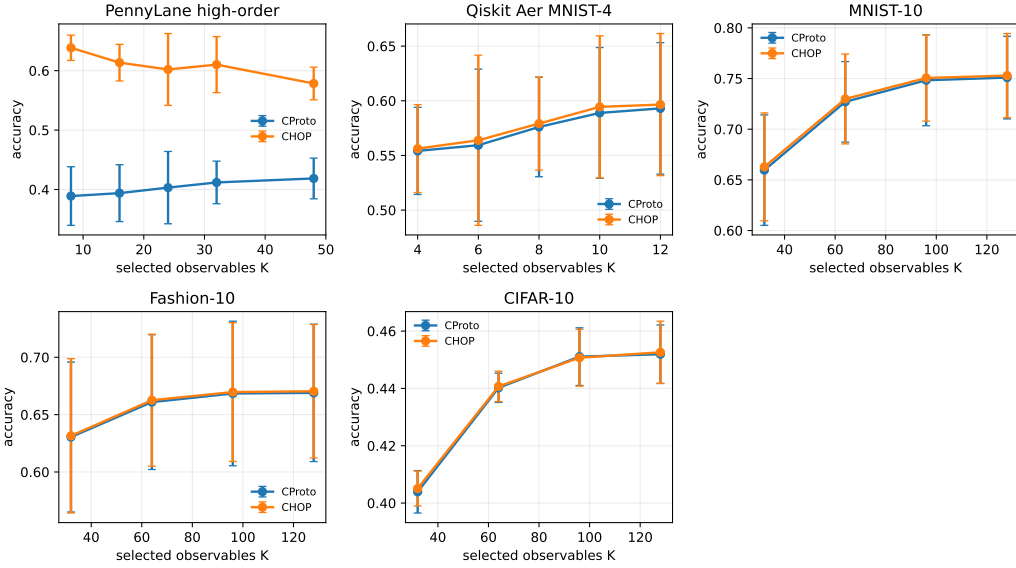


Figure 3: CHOP communication sweeps. CHOP and CProto use the same selected observable keys; the gap isolates mask-aware high-order statistics.

Table 23: Full paired significance tests for the main experimental claims. Reported values are mean paired accuracy deltas over matched seeds. One-sided paired t -tests test the directional claim; two-sided values are provided as a robustness check. Holm correction is applied across all reported comparisons.

Setting	Comparison	n	Δ acc.	d_z	Holm p_{1s}	Holm p_{2s}
MNIST-10	Masked-HOP - zero-fill	5	0.052	4.67	0.0059	0.0119
MNIST-10	Masked-HOP - shared-observable	5	0.050	4.78	0.0057	0.0113
MNIST-10	Masked-HOP - CProto	5	0.007	2.98	0.0250	0.0500
MNIST-10	CHOP - zero-fill	5	0.047	3.47	0.0149	0.0297
Fashion-10	Masked-HOP - zero-fill	5	0.015	1.86	0.0851	0.1702
Fashion-10	Masked-HOP - shared-observable	5	0.006	0.65	0.4422	0.8844
Fashion-10	Masked-HOP - CProto	5	0.003	1.72	0.0869	0.1739
Fashion-10	CHOP - zero-fill	5	0.013	1.53	0.1058	0.2116
CIFAR-10	Masked-HOP - zero-fill	5	0.033	8.16	0.0009	0.0018
CIFAR-10	Masked-HOP - shared-observable	5	0.029	1.83	0.0851	0.1702
CIFAR-10	Masked-HOP - CProto	5	0.004	2.18	0.0574	0.1148
CIFAR-10	CHOP - zero-fill	5	0.029	7.85	0.0010	0.0020
Covariance sensing	CHOP - zero-fill	5	0.105	3.54	0.0144	0.0289
Covariance sensing	CHOP - shared-observable	5	0.106	1.53	0.1058	0.2116
Covariance sensing	CHOP - CProto	5	0.374	10.35	0.0004	0.0007
Covariance sensing	Masked-HOP - Masked	5	0.319	8.40	0.0008	0.0016
MNIST-10	Masked-HOP - schema-FedProto	5	0.332	11.56	0.0003	0.0005
MNIST-10	Masked-HOP - schema-FedAdam	5	0.288	9.27	0.0006	0.0011
MNIST-10	Masked-HOP - no-schema	5	0.525	19.85	< 0.0001	< 0.0001
Fashion-10	Masked-HOP - schema-FedProto	5	0.140	6.17	0.0025	0.0050
Fashion-10	Masked-HOP - schema-FedAdam	5	0.118	5.04	0.0049	0.0099
Fashion-10	Masked-HOP - no-schema	5	0.423	23.67	< 0.0001	< 0.0001
CIFAR-10	Masked-HOP - schema-FedProto	5	0.186	10.49	0.0004	0.0007
CIFAR-10	Masked-HOP - schema-FedAdam	5	0.172	10.76	0.0003	0.0007
CIFAR-10	Masked-HOP - no-schema	5	0.288	39.71	< 0.0001	< 0.0001
PennyLane high-order	CHOP - CProto	3	0.199	10.72	0.0255	0.0511
PennyLane high-order	CHOP - schema-FedProto	3	0.317	25.23	0.0063	0.0126
Qiskit Aer	Masked-HOP - schema-FedProto	3	0.169	2.91	0.1058	0.2116
Qiskit Aer	Masked-HOP - FedDyn	3	0.178	3.48	0.1058	0.2116
UCI HAR	CProto eq. - zero-fill RFF eq.	5	0.409	13.34	0.0002	0.0003
UCI HAR	CProto eq. - shared-view RFF eq.	5	0.362	24.84	< 0.0001	< 0.0001
UCI HAR	Coverage all - anchor AE	5	0.028	5.38	0.0040	0.0079
MFeat	Group-CProto eq. - zero-fill RFF eq.	10	0.110	1.89	0.0031	0.0062
MFeat	Group-CProto eq. - shared-view RFF eq.	10	0.182	4.17	< 0.0001	< 0.0001
MFeat	Group-CProto all - mask-aware MLP	10	0.008	1.58	0.0083	0.0166
MHEALTH	CHOP eq. - CProto eq.	10	0.005	0.09	1.0000	1.0000
MHEALTH	CHOP eq. - zero-fill RFF eq.	10	0.091	1.05	0.0579	0.1158
MHEALTH	CHOP eq. - shared-view RFF eq.	10	0.145	1.27	0.0255	0.0511
MHEALTH	CHOP eq. - mask-aware MLP	10	0.005	0.07	1.0000	1.0000
MHEALTH	Adaptive CHOP - CProto eq.	10	0.018	0.92	0.0869	0.1739
MHEALTH	Random HOP - random CProto eq.	10	0.000	0.09	1.0000	1.0000
UCI HAR	Adaptive CHOP - CHOP eq.	5	0.120	4.82	0.0057	0.0113
MFeat	Adaptive CHOP - CHOP eq.	10	0.144	1.62	0.0071	0.0142
PAMAP2	Adaptive CHOP - CHOP eq.	5	0.230	2.74	0.0269	0.0537
PAMAP2	Adaptive CHOP - zero-fill RFF eq.	5	0.116	2.92	0.0255	0.0511
PAMAP2	Adaptive CHOP - shared-view RFF eq.	5	0.450	16.60	< 0.0001	0.0001

Table 24: High-order readout experiments. The PennyLane benchmark uses circuit-generated readouts with class means equalized, so the dominant signal is second-order.

Setting	Method	Acc.	95% CI	Bytes
PennyLane high-order	QProto-Masked-HOP	0.550	0.009	3476
PennyLane high-order	CHOP, $K = 24$	0.602	0.061	2324
PennyLane high-order	QProto-Masked	0.422	0.044	1940
PennyLane high-order	CProto, $K = 24$	0.403	0.061	788
PennyLane high-order	shared-observable	0.446	0.053	1556
PennyLane high-order	old QProto-HOP	0.320	0.028	3092
PennyLane high-order	no-schema	0.338	0.057	1556
Synthetic covariance	QProto-Masked-HOP	0.808	0.083	4116
Synthetic covariance	QProto-Masked	0.467	0.085	2068
Synthetic covariance	old QProto-HOP	0.795	0.059	2260
Synthetic covariance	low-order prototype	0.449	0.115	212
Synthetic covariance	wrong-schema HOP	0.445	0.029	2260

Table 25: CHOP communication sweeps. CProto uses the same selected observable keys but omits HOP.

Setting	Variant	K	Acc.	Bytes
PennyLane high-order	CProto	8	0.389	276
PennyLane high-order	CHOP	8	0.639	1812
PennyLane high-order	CHOP	24	0.602	2324
PennyLane high-order	CHOP	48	0.579	3092
MNIST-10	CHOP	32	0.663	6444
MNIST-10	CHOP	64	0.730	9004
MNIST-10	CHOP	96	0.751	11564
MNIST-10	CHOP	128	0.753	14124
Qiskit Aer MNIST-4	CHOP	4	0.556	404
Qiskit Aer MNIST-4	CHOP	8	0.579	532
Qiskit Aer MNIST-4	CHOP	10	0.594	596
Qiskit Aer MNIST-4	CHOP	12	0.597	660

Table 26: CHOP key-selection controls. Anchor-informed key policies are compared to random keys at the same K .

Setting	Key policy	Method	Acc.	Bytes
PennyLane high-order, $K = 8$	coverage	CHOP	0.659	1812
PennyLane high-order, $K = 8$	variance	CHOP	0.624	1812
PennyLane high-order, $K = 8$	random	CHOP	0.431	1812
PennyLane high-order, $K = 8$	coverage	CProto	0.394	276
MNIST-10, $K = 96$	variance	CHOP	0.751	11564
MNIST-10, $K = 96$	coverage	CHOP	0.749	11564
MNIST-10, $K = 96$	random	CHOP	0.662	11564
CIFAR-10, $K = 96$	coverage	CHOP	0.452	11564
CIFAR-10, $K = 96$	variance	CHOP	0.451	11564
CIFAR-10, $K = 96$	random	CHOP	0.391	11564
Qiskit Aer, $K = 10$	variance	CHOP	0.594	596
Qiskit Aer, $K = 10$	coverage	CHOP	0.590	596
Qiskit Aer, $K = 10$	random	CHOP	0.582	596

Table 27: Paired seed-level accuracy deltas for the main high-order and key-selection claims.

Comparison	n	Mean delta	Wins
Masked-HOP minus Masked, PennyLane high-order	3	0.128 ± 0.042	3/3
Masked-HOP minus old HOP, PennyLane high-order	3	0.230 ± 0.035	3/3
CHOP minus CProto, PennyLane high-order $K = 8$	3	0.250 ± 0.071	3/3
Coverage-key CHOP minus random-key CHOP, $K = 8$	5	0.228 ± 0.066	5/5
Variance-key CHOP minus random-key CHOP, MNIST-10	3	0.089 ± 0.016	3/3
Coverage-key CHOP minus random-key CHOP, CIFAR-10	3	0.061 ± 0.044	3/3
CHOP minus schema-FedProto, MNIST-10	3	0.333 ± 0.027	3/3
CHOP minus schema-FedProto, CIFAR-10	3	0.179 ± 0.058	3/3
CHOP minus schema-FedProto, Qiskit Aer	3	0.151 ± 0.227	3/3
CHOP minus schema-FedProto, PennyLane high-order	3	0.317 ± 0.031	3/3
CHOP minus schema-FedAdam, MNIST-10	3	0.291 ± 0.031	3/3
CHOP minus schema-FedAdam, CIFAR-10	3	0.169 ± 0.054	3/3
CHOP minus schema-FedAdam, PennyLane high-order	3	0.286 ± 0.042	3/3

Table 28: Schema comparability benchmark on synthetic heterogeneous readout.

Method	Acc.	Std.	Bal. acc.
Full + shrinkage	0.621	0.055	0.620
No schema	0.422	0.047	0.421
Forced canonical	0.415	0.019	0.415
Wrong schema	0.290	0.010	0.292

Table 29: High-order outer-product benchmark on covariance-structured data.

Method	Acc.	Std.	Bal. acc.
Full + shrinkage	0.795	0.027	0.799
QProto-HOP	0.795	0.024	0.798
Prototype	0.449	0.046	0.449
Wrong schema	0.445	0.012	0.445
No schema	0.308	0.013	0.307

Table 30: CIFAR-4 shot-noise sweep. Results are means over 5 seeds.

Shots	Full	Proto	No schema	Wrong	FedAvg
32	0.474	0.469	0.334	0.261	0.286
64	0.476	0.471	0.336	0.261	0.286
128	0.475	0.472	0.337	0.262	0.287
256	0.474	0.472	0.337	0.262	0.287
1024	0.473	0.472	0.338	0.262	0.287

Table 31: Matched communication on the high-order synthetic benchmark.

Budget	Method	Bytes	Acc.
Low	Prototype	212	0.449
Low	HOP	212	0.504
High	Prototype	2260	0.749
High	HOP	2260	0.795

Table 32: CIFAR-4 matched communication curve.

Bytes	Prototype	HOP
1556	0.446	0.429
3092	0.488	0.446
4116	0.496	0.468

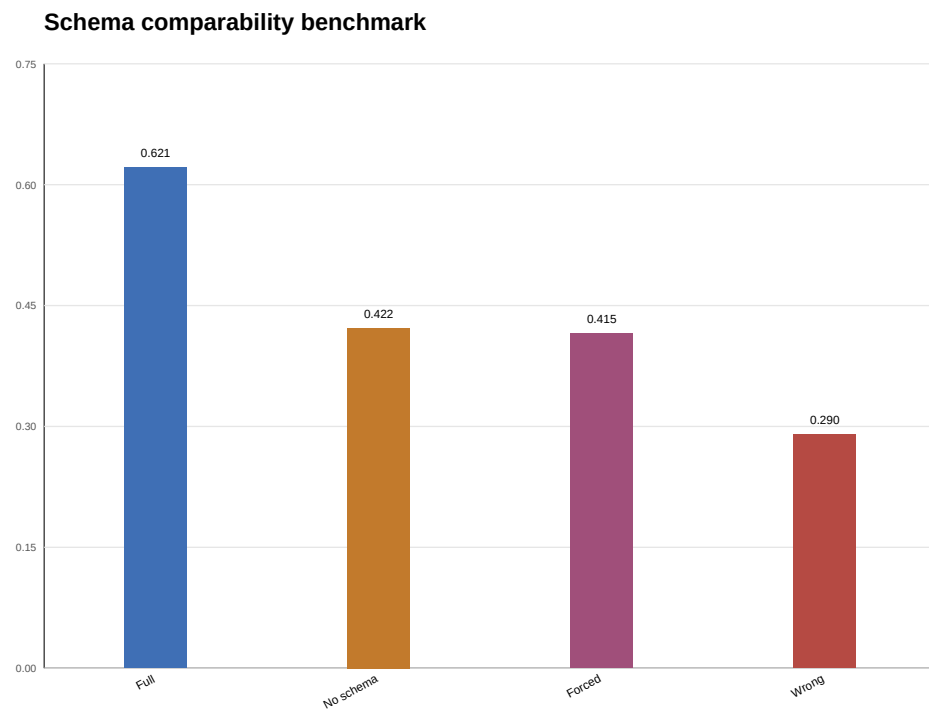


Figure 4: Schema comparability benchmark. The gain comes from making readout coordinates semantically comparable before aggregation.

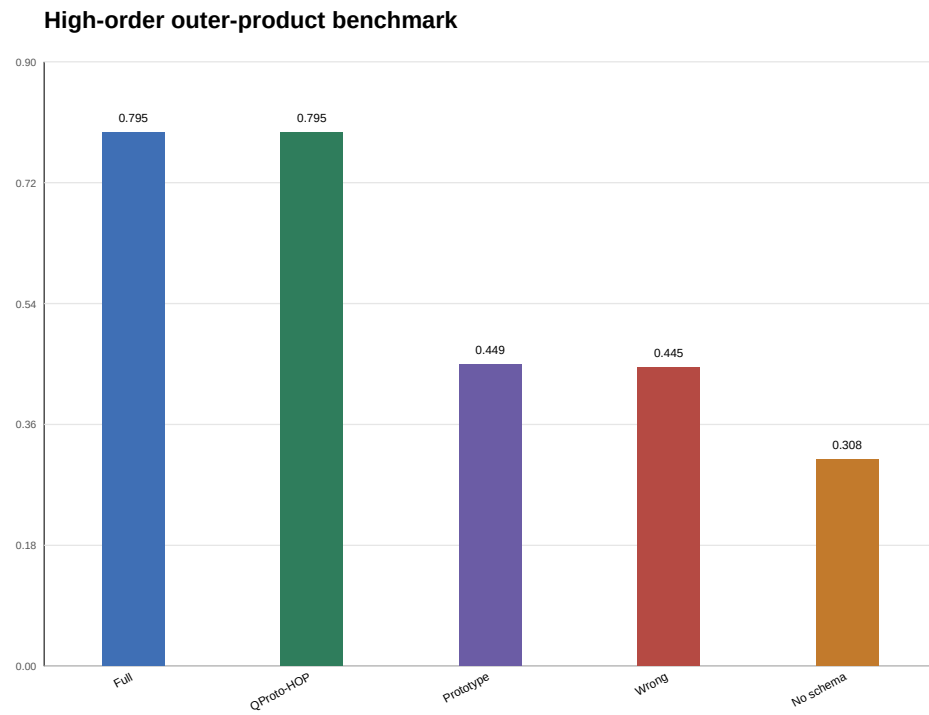


Figure 5: High-order benchmark. HOP separates covariance-structured classes that low-order prototypes collapse.

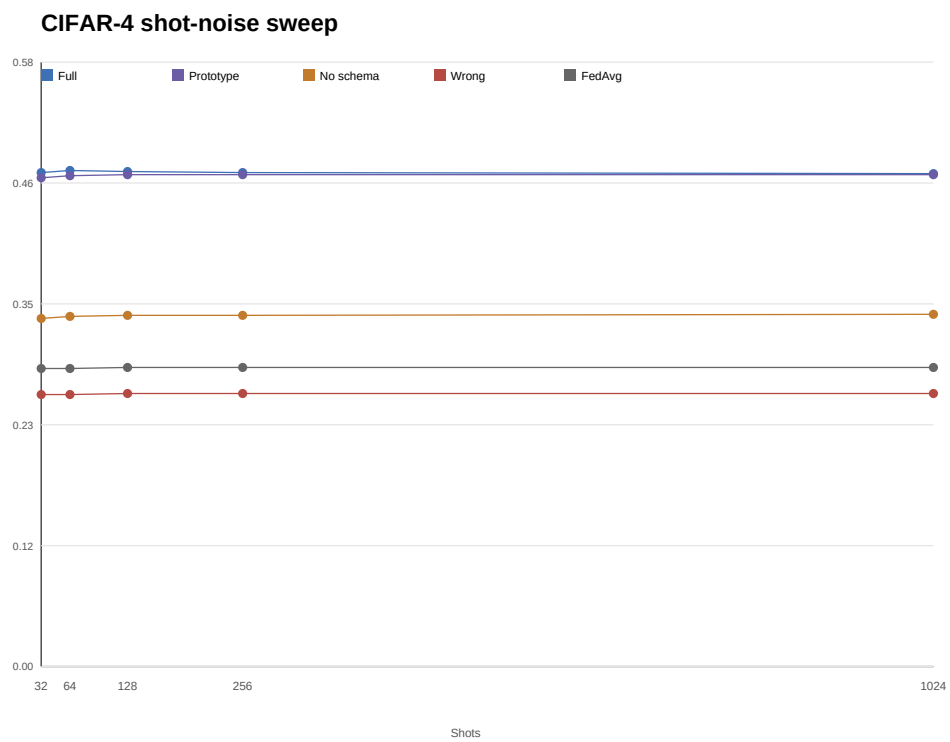


Figure 6: CIFAR-4 shot sweep under heterogeneous readout.

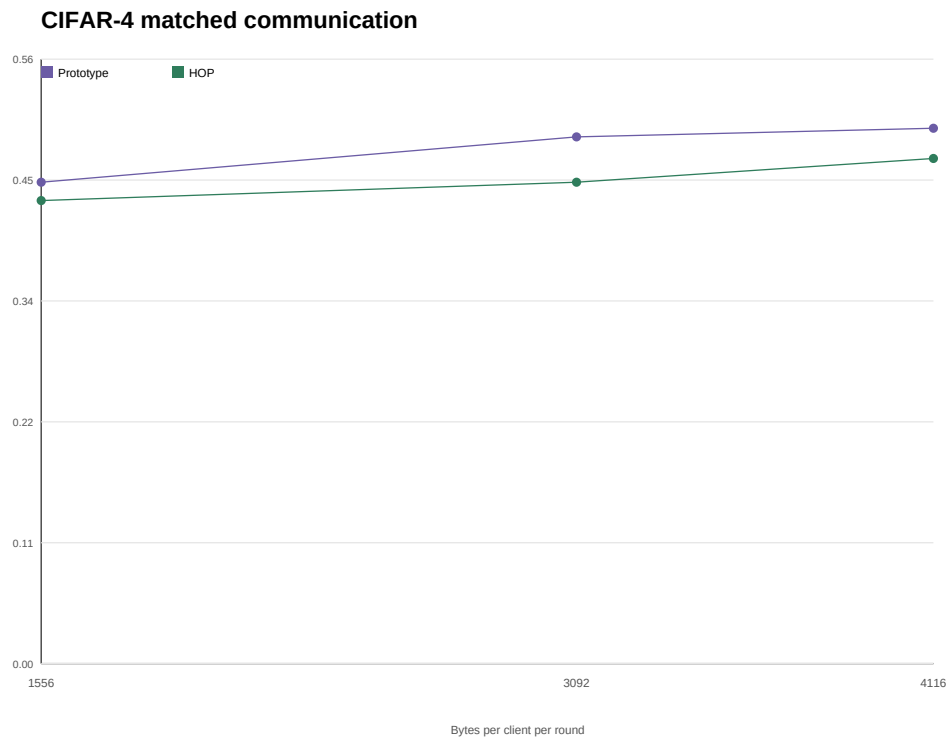


Figure 7: CIFAR-4 communication curve. On this low-order-separable real feature task, HOP is not the primary source of gain.